

Allampati Bharath Kumar Reddy

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ABOUT ME

CSE undergraduate with hands-on experience in coding, networking, and security tools. Driven to develop efficient, real-world applications using modern technologies. Quick learner with strong problem-solving skills and a passion for continuous growth.

EDUCATION

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|---|-------------------------|
| • Vellore Institute of Technology | 2022 – 2026 |
| B. Tech in Computer Science (Specialization: Information Security), 8.41/10 CGPA | Vellore, Tamil Nadu |
| • Narayana Junior College | 2020 – 2022 |
| Class 12(State Board), 92.2% Higher Secondary | Nellore, Andhra Pradesh |
| • Narayana High School | 2019 – 2020 |
| Class 10(State Board), 99.5 % | Nellore, Andhra Pradesh |

PROJECTS

Digital Watermarking System([link](#))

Python, Image Processing, LSB, DCT, QR Encoding, ML

- Implemented a robust digital watermarking framework supporting both LSB-based and DCT-based embedding techniques for securely hiding QR-encoded data inside color images.
- Designed end-to-end watermark workflows including QR generation, bit-level encoding, block-wise DCT coefficient manipulation, and image reconstruction with minimal perceptual distortion.
- Achieved reliable watermark extraction even under transformations such as compression, noise addition, and format changes, demonstrating strong resilience and high PSNR accuracy.

Smart-Queue-Alert([link](#))

Cloud Computing, AI, Image Processing, AWS

- Designed and executed a cloud-based crowd detection system using **AWS Rekognition** integrated with **Google Colab** for real-time image analysis.
- Automated the detection and counting of people in uploaded canteen queue images, classifying queue levels as *Low*, *Moderate*, or *Crowded* using predefined thresholds.
- Utilized **Amazon S3** for image storage and **IAM-secured access** for Rekognition API calls, ensuring a secure and fully cloud-native workflow.

Quiz-Me([link](#))

JavaScript, Node.js, Web Development

- Engineered an interactive MCQ quiz platform with real-time scoring, dynamic question rendering, and instant feedback mechanisms to enhance user engagement.
- Enforced modular question management enabling scalable quiz sets and consistent evaluation logic across topics.
- Optimized the application workflow, improving load efficiency and contributing to a smoother user experience.

SKILLS

- **Languages:** C, Java, SQL, Python
- **Core CS Fundamentals:** Data Structures & Algorithms, Object Oriented Programming, Operating Systems, Database Management Systems, Computer Networks
- **Tools & Platforms:** Git, GitHub, Visual Studio Code, PostgreSQL, MySQL, Wireshark
- **Cloud Devops :** AWS(EC2,S3,Lambda,Rekognition,API Gateway, IAM, CLI)

CERTIFICATES AND TRAINING

Oracle OCI Generative AI Professional ([link](#))

May2025 – June 2025

- Gained hands-on experience with SIEM, IDS, threat detection, and Python scripting.

Google Cybersecurity Professional Certificate ([link](#))

Sep 2024 – Feb 2025

- Completed the Google Cybersecurity Professional Certificate, gaining hands-on experience in threat detection, risk mitigation, incident response, and security operations using industry-standard tools.

EXTRACURRICULAR ACTIVITIES

- Member of **Way of Living Chapter**, focused on personal growth and value-based living. Participated in sessions on mindfulness, discipline, and holistic development.
- Online Internship on “Machine Learning using Python” at Pantech Solutions, hands-on experience in ML algorithms and mini-projects.